ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C3 · ATOMS

Parts of an atom

★☆☆
Grade 4

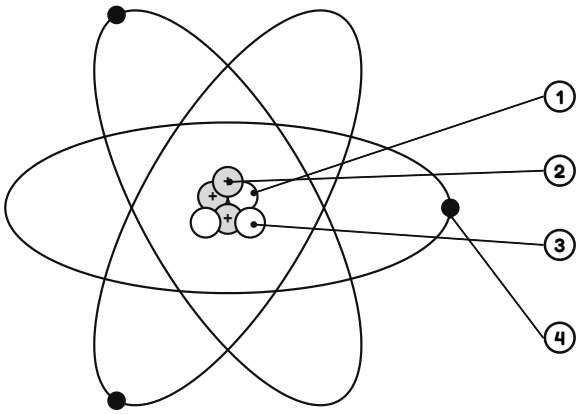
Learning goals

- Label an atom.
- Recall charges.

1 Label the atom. [4]

LABELS

neutron nucleus electron proton



(1) nucleus (2) proton

(3) neutron (4) electron

2 Complete the table. [6]

Particle	Charge	Where?
proton	positive (+)	nucleus
neutron	none	nucleus
electron	negative (–)	around the nucleus

3 Fill in the gaps. [4]

WORD BANK

neutron protons nucleus electron cells atoms

All matter is made of atoms. The centre of an atom is the nucleus. The number of protons decides which element an atom is. The lightest particle is the electron.

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. Neutrons have no charge.	✔		
2. Electrons are heavier than protons.		✔	Electrons are about 2000 times lighter.
3. There are about 118 kinds of atom.	✔		
4. An atom is mostly empty space.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C3 · ATOMS

Thinking about atoms

★★☆
Grade 4

Learning goals

- Use the atom model.
- Compare particles.

1 **An atom of nitrogen has 7 protons. How many electrons does it have? Explain.** [2]

7 electrons, because atoms have the same number of protons and electrons so the charges balance.

2 **Most of the mass of an atom is in the nucleus. Explain why.** [2]

Protons and neutrons are heavy and are in the nucleus; electrons are very light.

3 **Two atoms both have 6 protons. Are they the same element? Explain.** [2]

Yes: both are carbon, because the number of protons decides the element.

4 **Give one way the drawing of an atom on the Learn it page is useful and one way it is not like a real atom.** [2]

Useful: it shows the nucleus, protons, neutrons and electrons and where they are. Not like a real atom: the nucleus is drawn far too big and electrons do not move in neat circles.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C3 · ATOMS

Build and compare atom models

★★☆
Grade 4

Learning goals

- Make models from data.
- Find patterns in data.
- Evaluate models.

You will need

modelling clay or paper circles in three colours (one for protons, one for neutrons, one for electrons), card, glue.

Make models of these atoms. Use the data to count the particles.

1 **Complete the table.** [8]

Element	Protons	Neutrons	Electrons	Total particles	Mass (units)
helium	2	2	2	6	4
lithium	3	4	3	10	7
carbon	6	6	6	18	12
oxygen	8	8	8	24	16

Mass = protons + neutrons (electrons are so light we leave them out).

2 **What pattern do you notice between the number of protons and the number of electrons?** [1]

They are always the same.

3 **Why do we not count the electrons when we work out the mass?** [1]

Electrons are about 2000 times lighter than protons, so they add almost nothing to the mass.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C3 · ATOMS

How did we find out about atoms?

★★★
Grade 4

Learning goals

- Learn how ideas change with new evidence.
- Order events on a timeline.
- Think about creativity in science.

Science | History | Maths

Nobody has ever seen inside an atom with their eyes. So how do we know what atoms are like? Over hundreds of years, scientists did clever experiments and changed their models when they found new **evidence**.

In **1803**, **John Dalton** in England said that each element is made of its own kind of atom, like tiny hard balls. In **1897**, **J. J. Thomson** discovered the **electron**, so atoms could not be solid balls. In **1911**, **Ernest Rutherford** fired tiny particles at a thin sheet of gold. Most went straight through, but a few bounced back! This showed that atoms are mostly empty space with a tiny, heavy **nucleus**. In **1932**, **James Chadwick** discovered the **neutron**.

Based on the history of atomic models.

1 **Put the discoveries in order on the timeline.** [4]

5	Chadwick discovers the neutron (1932)
1	Democritus names the atom (about 400 BCE)
4	Rutherford discovers the nucleus (1911)
2	Dalton: elements are made of atoms (1803)
3	Thomson discovers the electron (1897)

2 **Maths. How many years passed between Dalton's idea and Chadwick's discovery?** [1]

1932 − 1803 = 129 years

3 **Rutherford was surprised when some particles bounced back. Why did this make him change the model?** [2]

If atoms were solid balls or spread-out charge, the particles would not bounce back; the bounce showed there must be a tiny, heavy centre (the nucleus).

4 **Scientists combine evidence with creative thinking. Explain what this means using the story of the atom.** [2]

They did experiments to get evidence (like Rutherford's gold sheet), then imagined new models (a nucleus with electrons around it) to explain what they saw.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C3 · ATOMS

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

F	B	B	R	D	A	T	O	M	J
E	P	E	M	I	F	L	P	A	J
L	R	L	O	S	O	C	N	W	G
E	O	E	D	U	C	H	E	H	L
C	T	M	E	A	D	A	U	M	O
T	O	E	L	H	M	R	T	G	M
R	N	N	T	O	A	G	R	Y	N
O	W	T	K	H	S	E	O	D	R
N	D	B	G	S	S	W	N	A	B
S	M	N	U	C	L	E	U	S	W

atom charge electron element mass model neutron nucleus proton

2 Which particle has a negative charge? [1]

○ proton
○ neutron
● **electron**
○ nucleus

3 Which two particles are found in the nucleus? [1]

○ protons and electrons
● **protons and neutrons**
○ neutrons and electrons
○ only electrons

4 Draw a model of a helium atom (2 protons, 2 neutrons, 2 electrons). Label each part. [3]

Helium atom

Nucleus with 2 protons (+) and 2 neutrons; 2 electrons (–) around it; labels.

5 **Challenge. Explain why an atom has no overall charge even though it contains charged particles.** [2]

It has the same number of positive protons and negative electrons, so the charges cancel out.